1. Setup Your Project

Epi-Quant

Epi-Quant

1. Setup | 2. Extract | 3. Classify | 4. Correct Classification | 5. Track | 6. Correct Tracking | 7. Trajectory | 8. Data Playground | Clipboard | Light

Change the UI theme

Setup

Initialize your project folder

Store your images and masks in two folders

Masks folder: /example/masks (.tiff, .npy, .txt, …) Browse...

Images folder: /example/images (.png, .tiff, .jpg, …) Browse...

Output folder: /example/ Browse...

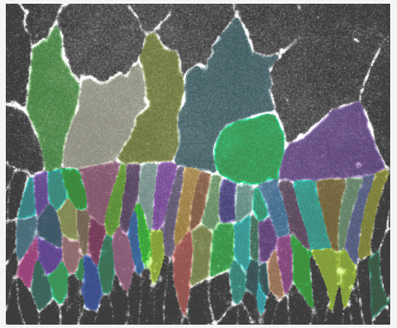
It will create three directories:

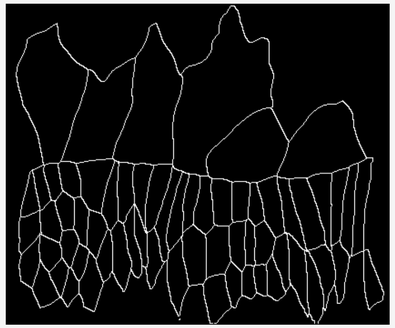
```
images
masks
cell_classification_output
cell_track_output
quantitative_analysis_output
```

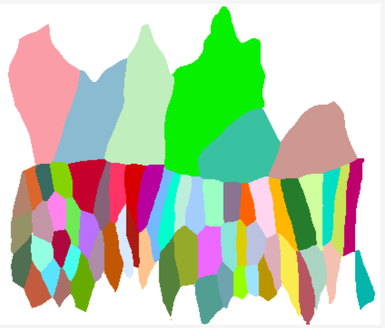
Set Project

Supporting mask file types

Cellpose mask (.npy) | EpySeg/TA binary image (.tif) | MorpholibJ label image (.tif)

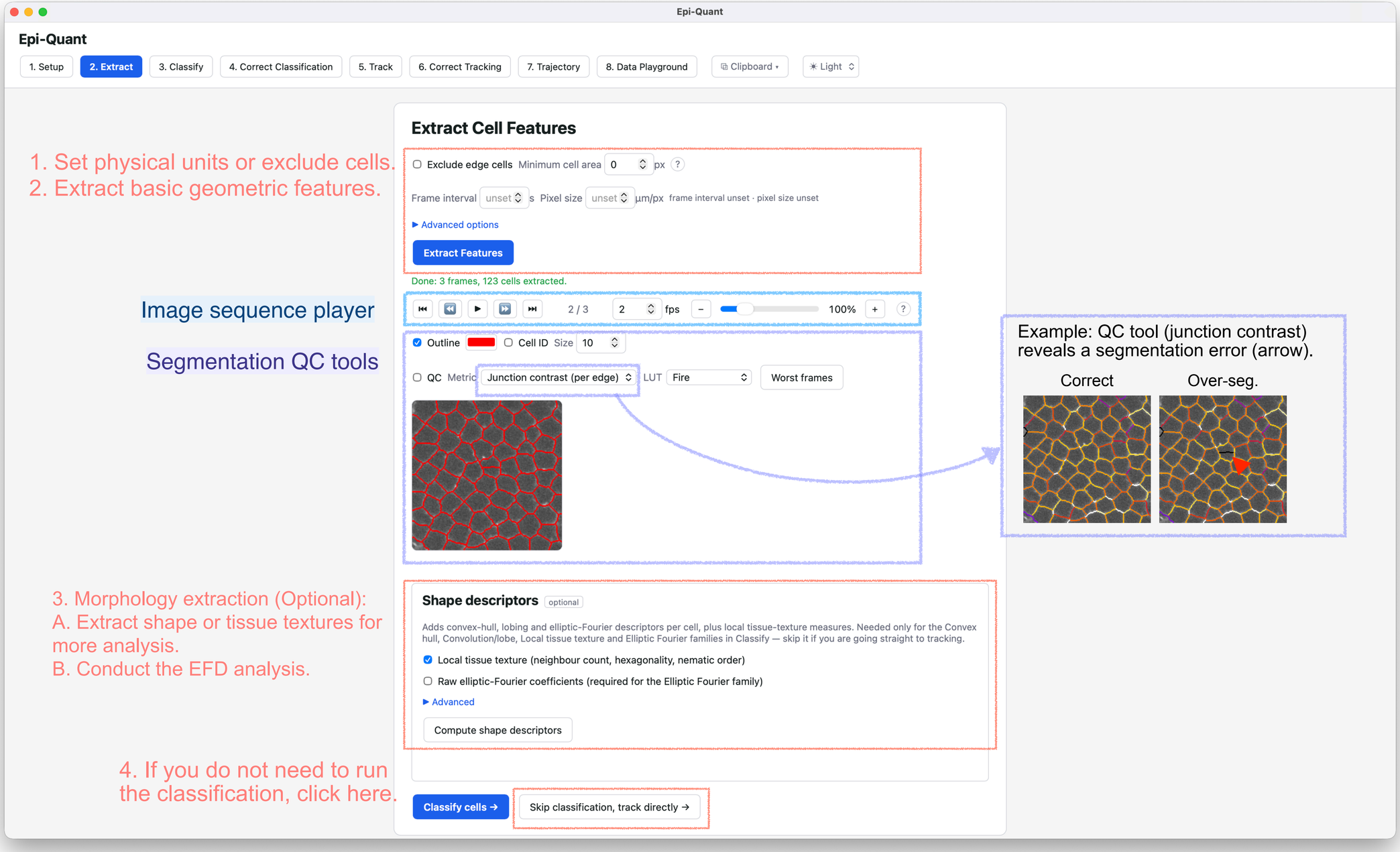




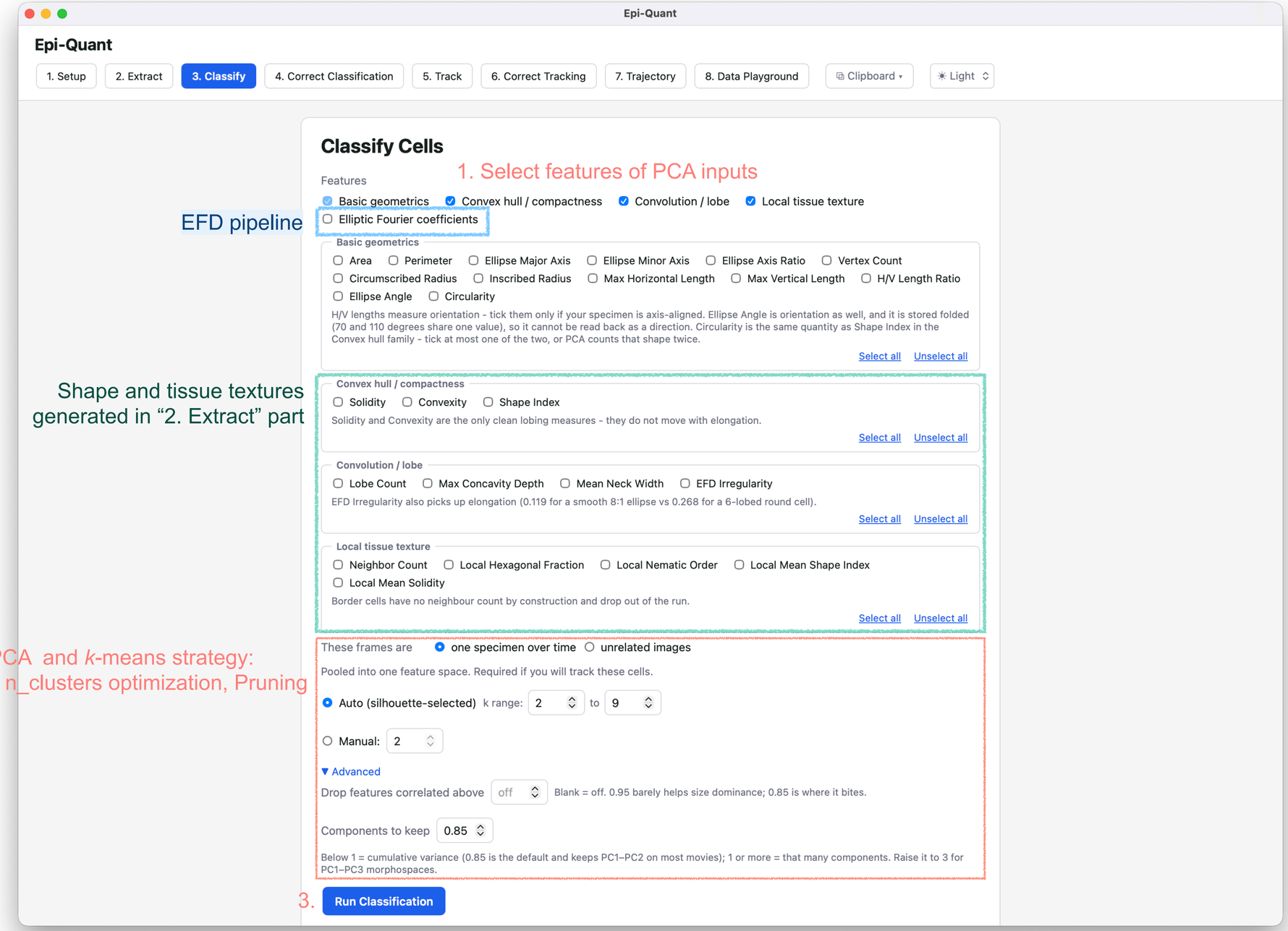


…

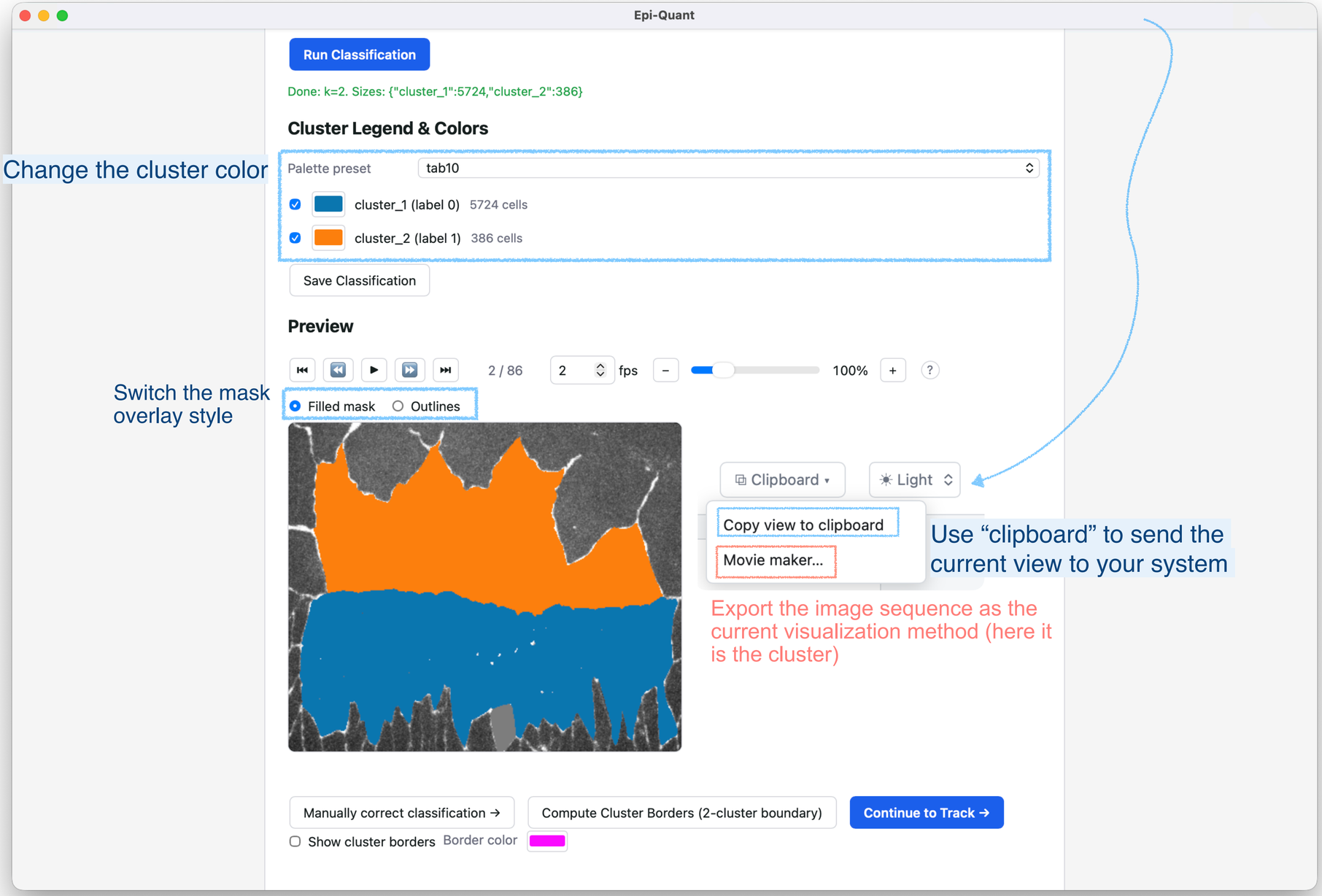
2. Segmentation QC and Feature Extraction for Further Analysis



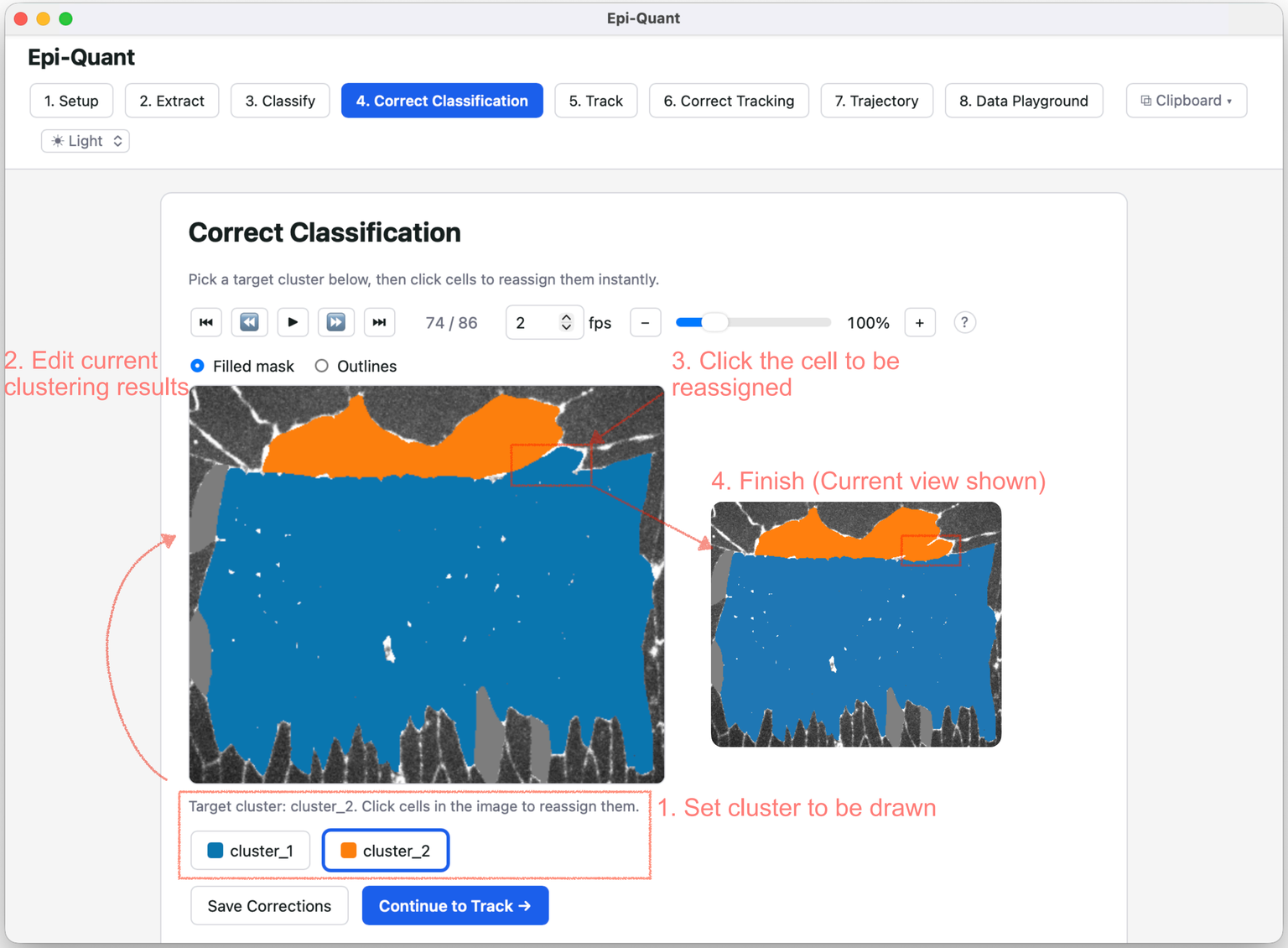
3.1 Dimension Reduction and Clustering Setup



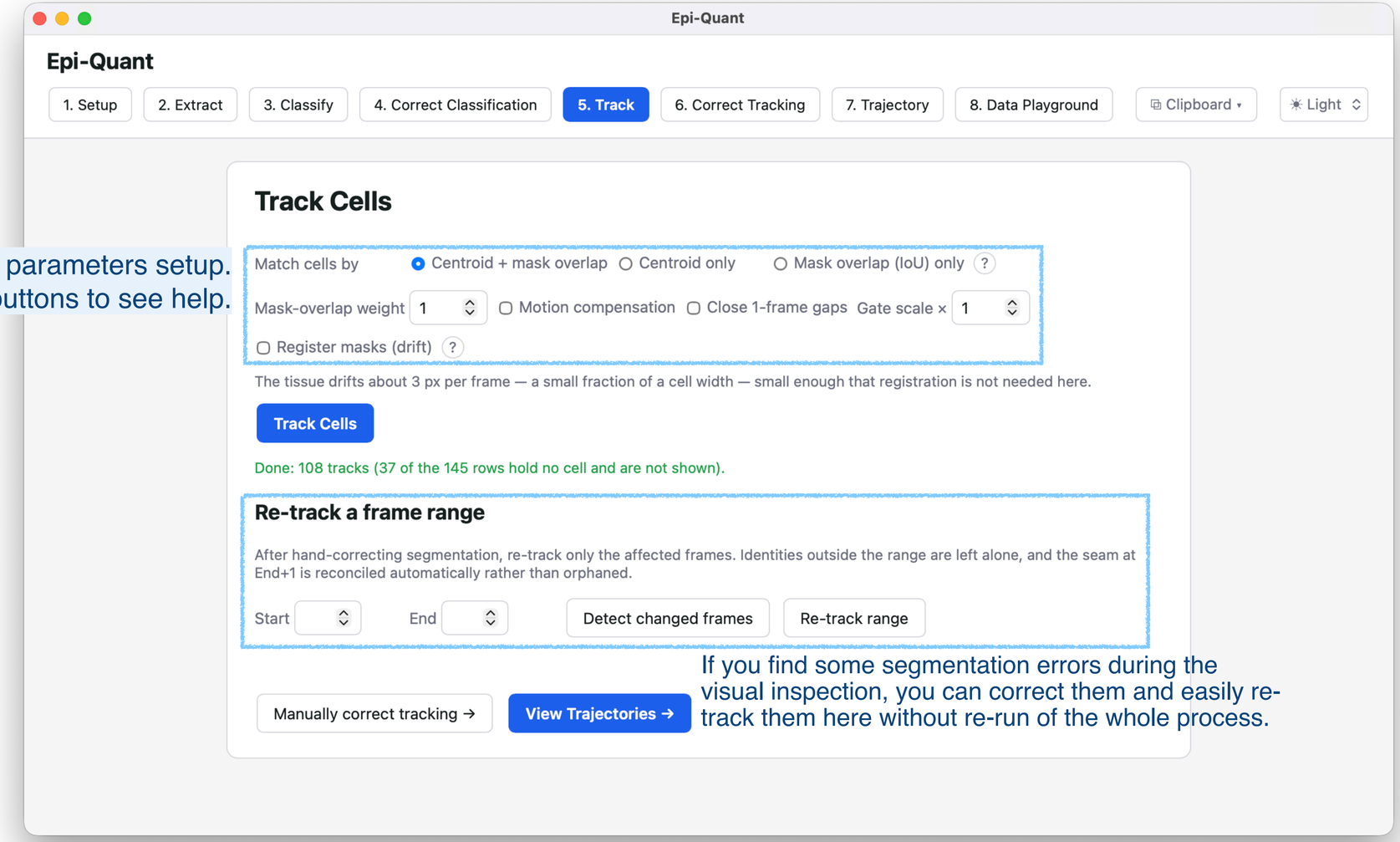
3.2 Preview of Classification Results



4. Manual Correction of Classification Results



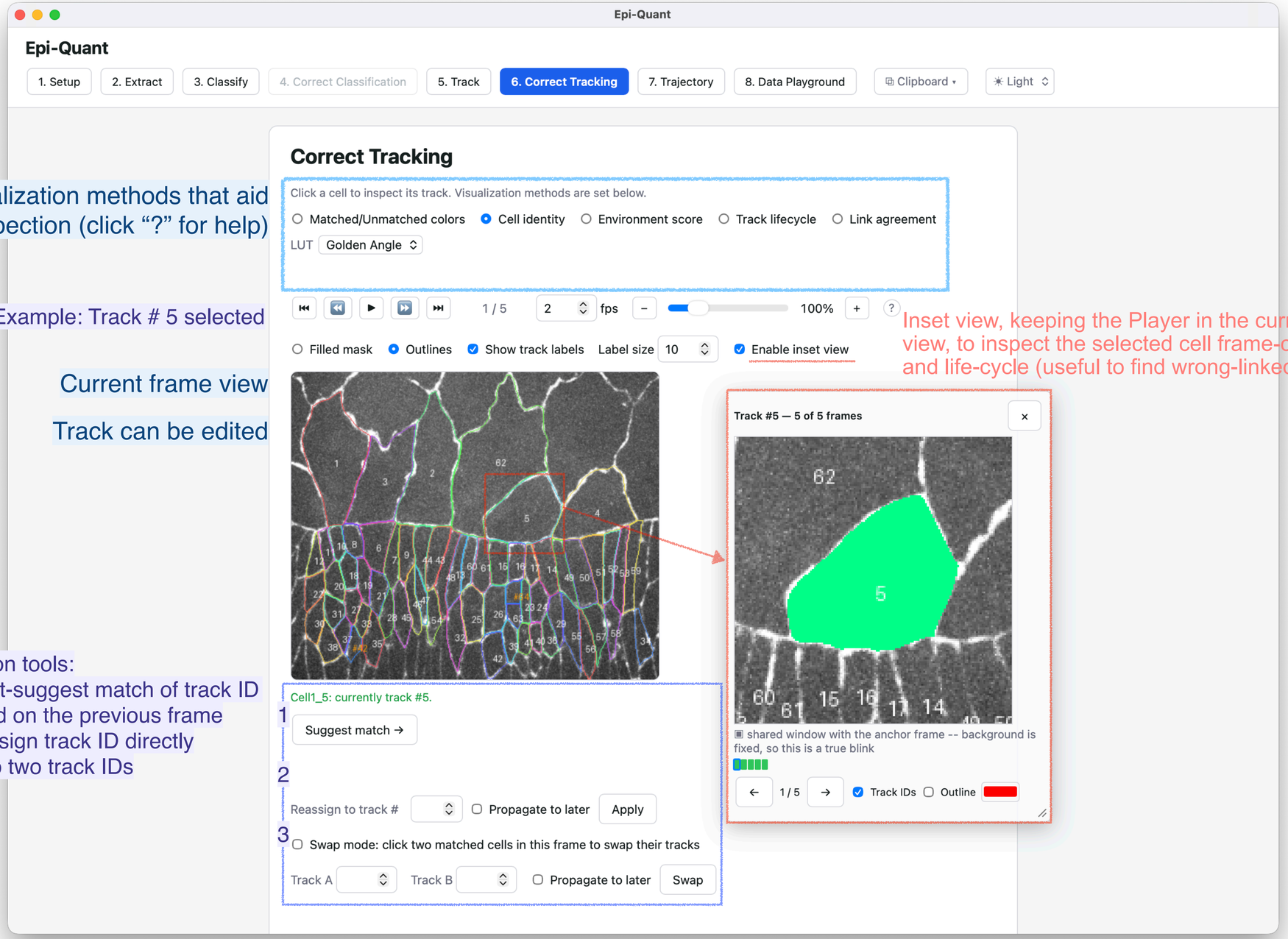
5. Track Setup and Local Re-track



Tracking parameters setup.
Click “?” buttons to see help.

If you find some segmentation errors during the visual inspection, you can correct them and easily re-track them here without re-run of the whole process.

6.1 Overview of Manual Tracking Correction Module



Visualization methods that aid visual inspection (click “?” for help)

Example: Track # 5 selected

Current frame view

Track can be edited

Inset view, keeping the Player in the current view, to inspect the selected cell frame-change and life-cycle (useful to find wrong-linked tracks)

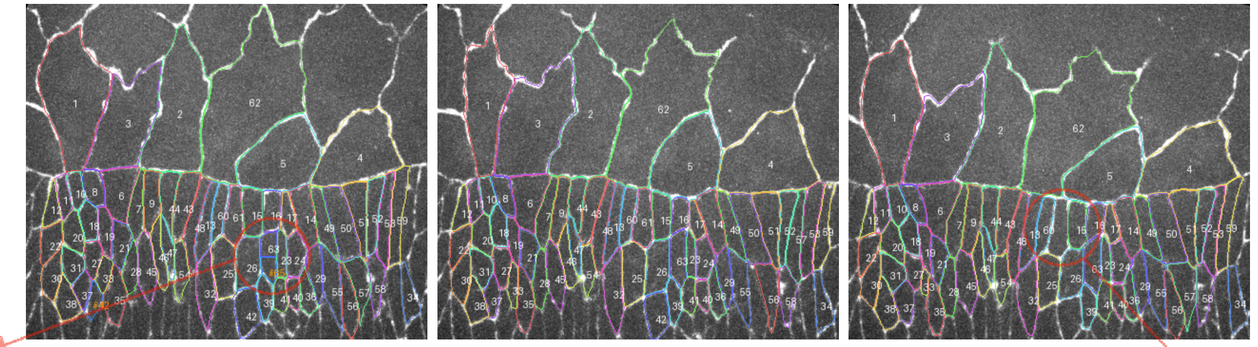
Correction tools:

1. Smart-suggest match of track ID based on the previous frame
2. Reassign track ID directly
3. Swap two track IDs

6.2 Example: Visualization Methods Enable The Segmentation QC After A Pre-Track

Track ID (color-encoded) view

Frame, *f* | *f*+1 | *f*+2 | *f*+3

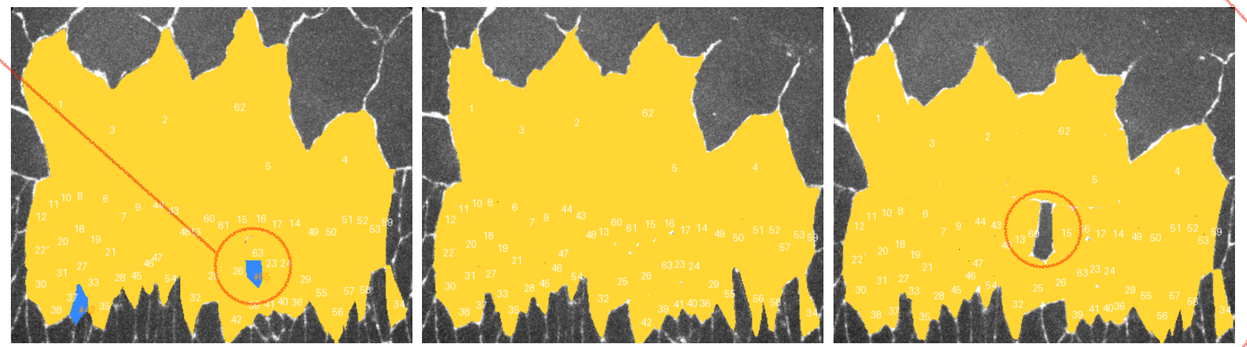


A over-segmentation error hinted by the unmatched color (unlinked during a pre-track)

Matched/Unmatched Track view

Matched = Track is linked to the next frame

Unmatched = A segmentation error or an orphan track

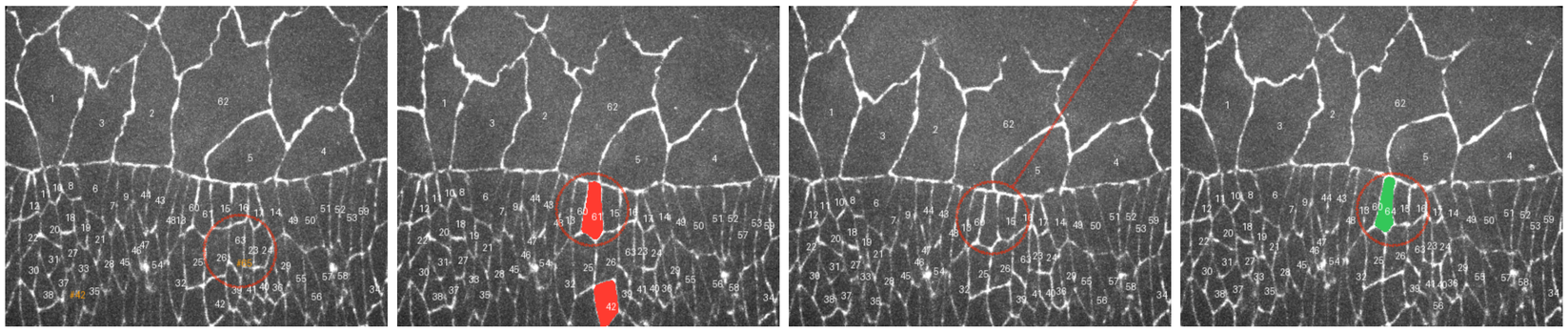


A under-segmentation error hinted by the life-cycle view (red-to-green sparkling track)

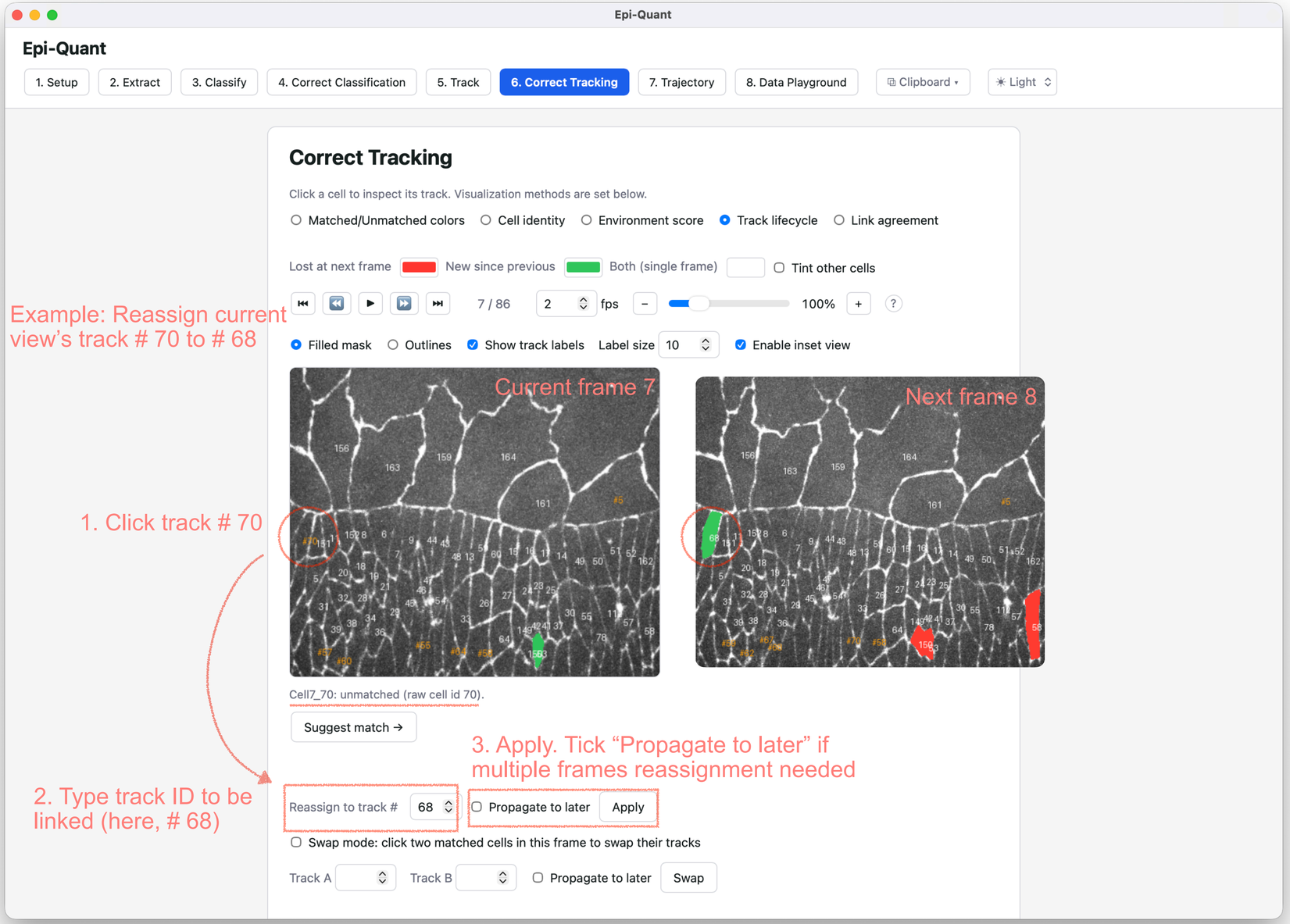
Track life-cycle view

Track lost at next frame

New track since previous

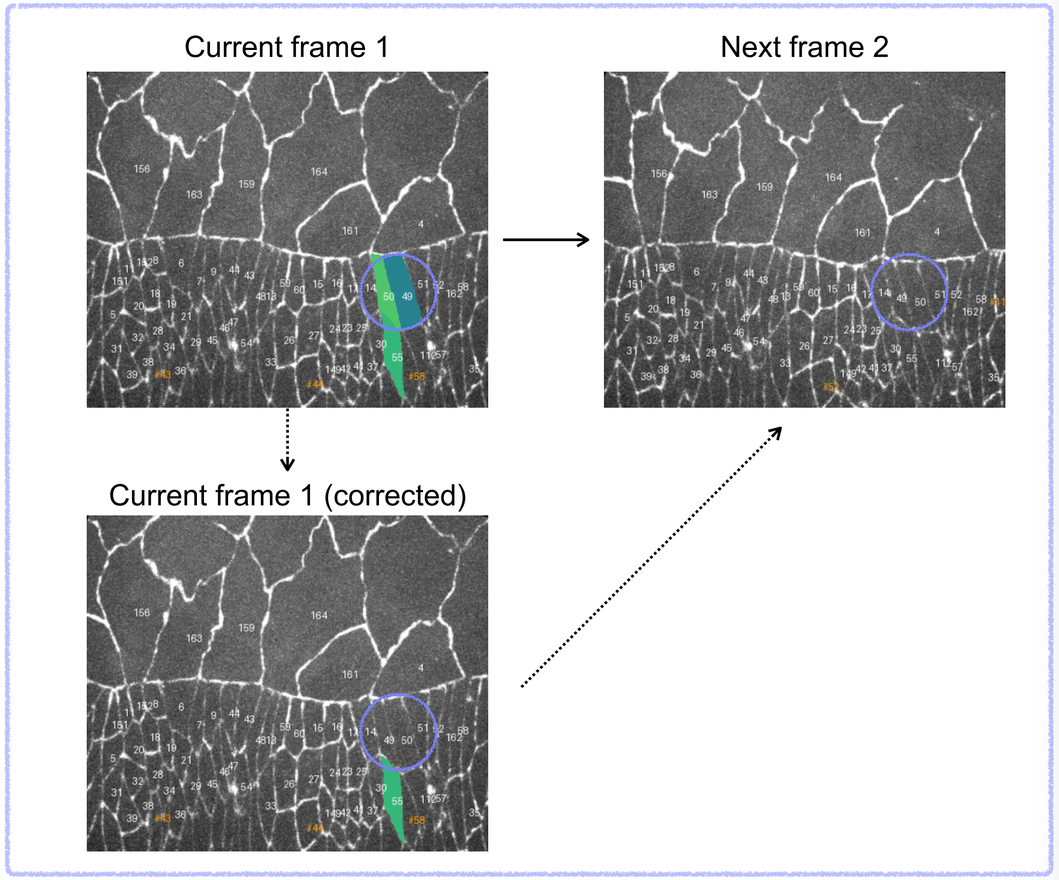


6.3 Tracking Correction by Reassigning Track Directly



6.4 Tracking Correction by Swapping Tracks

Example: Environment score (a non-reference metric) hints a track swapping error

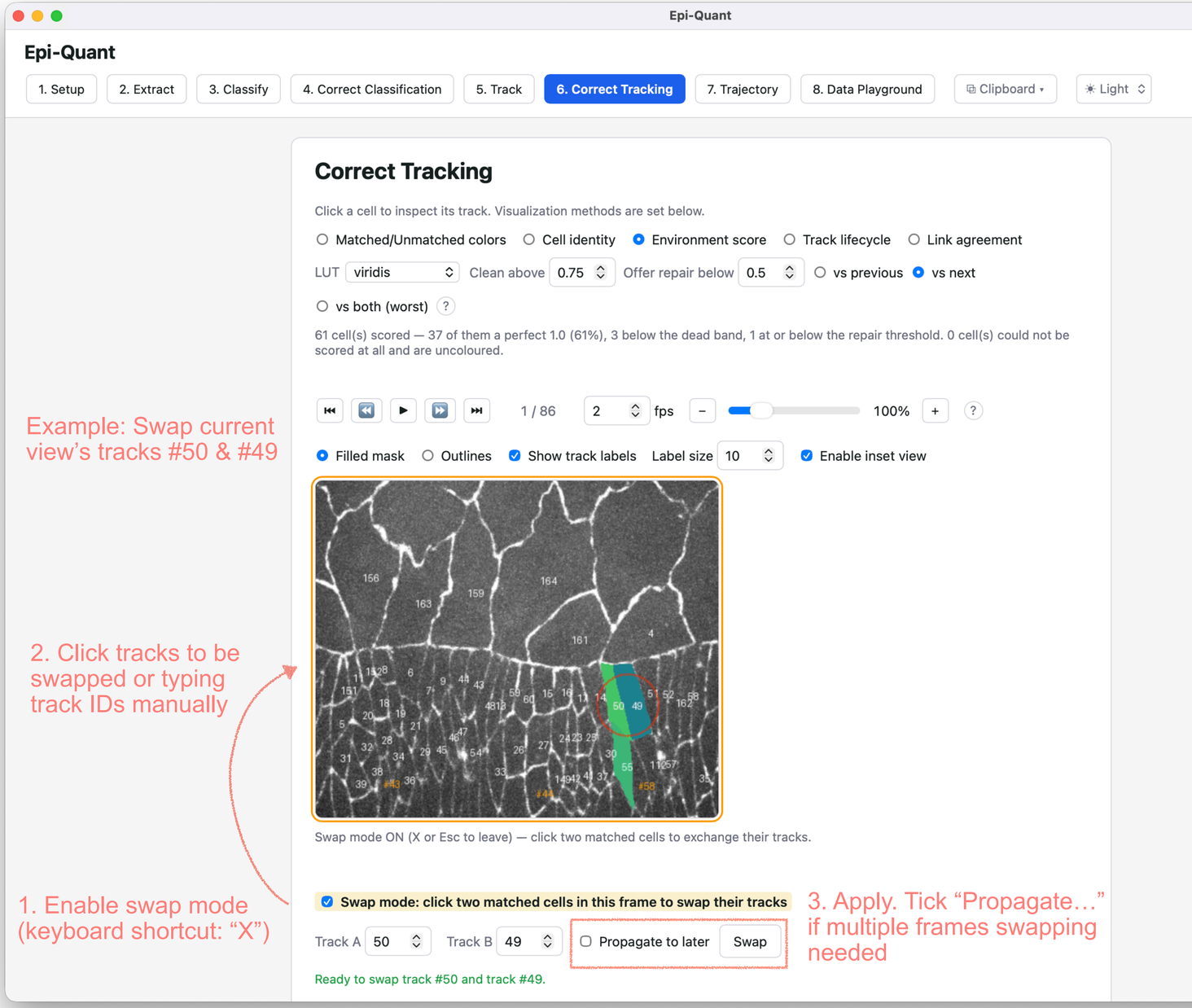


Example: Swap current view's tracks #50 & #49

2. Click tracks to be swapped or typing track IDs manually

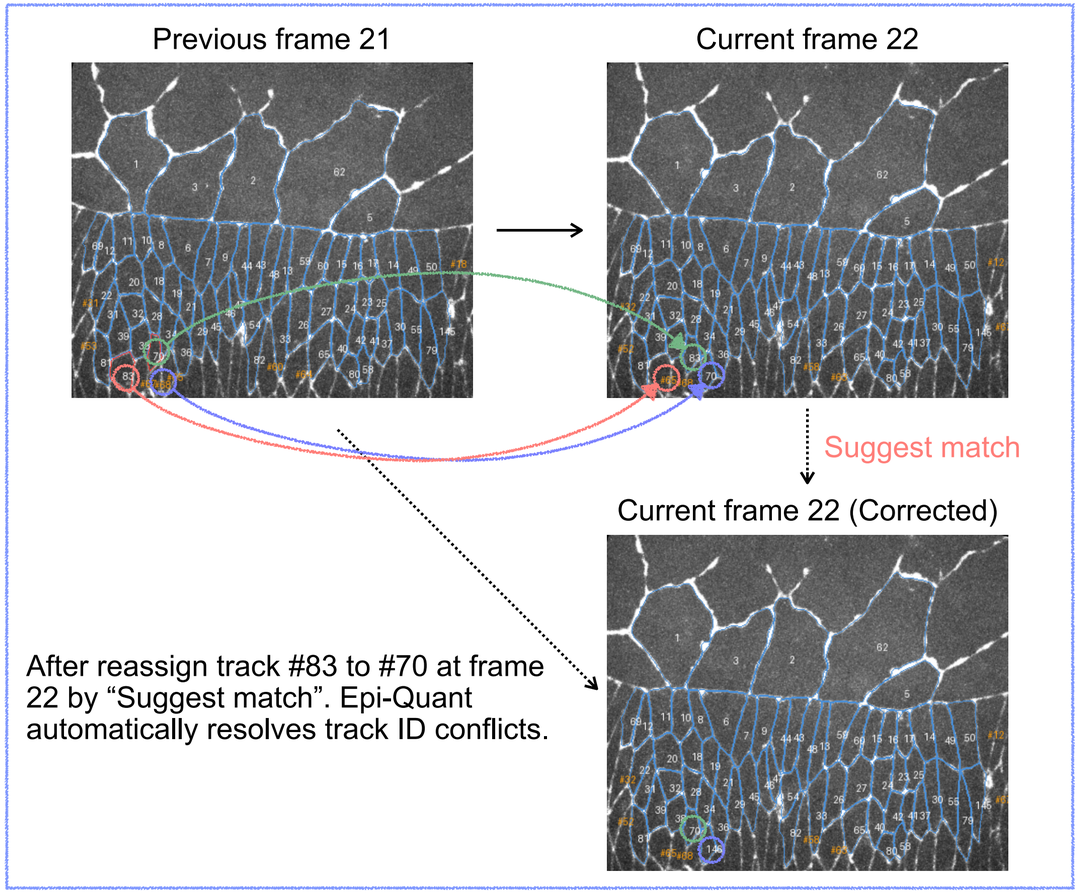
1. Enable swap mode (keyboard shortcut: "X")

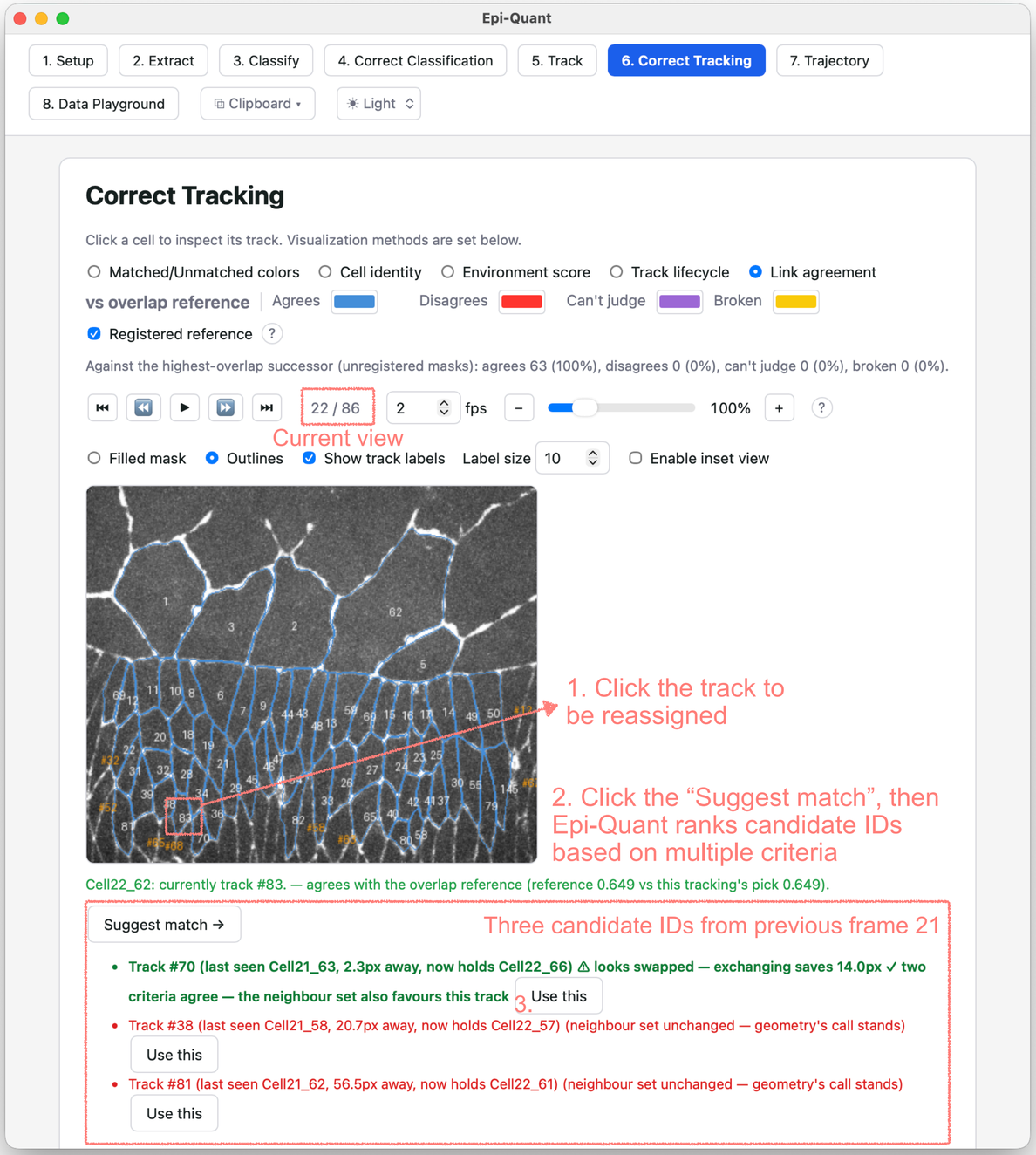
3. Apply. Tick "Propagate…" if multiple frames swapping needed



6.5 Tracking Correction with The Aid of Suggest Match

Example: Linkage agreement (an IoU-based, non-reference metric) hints a complex tracking error





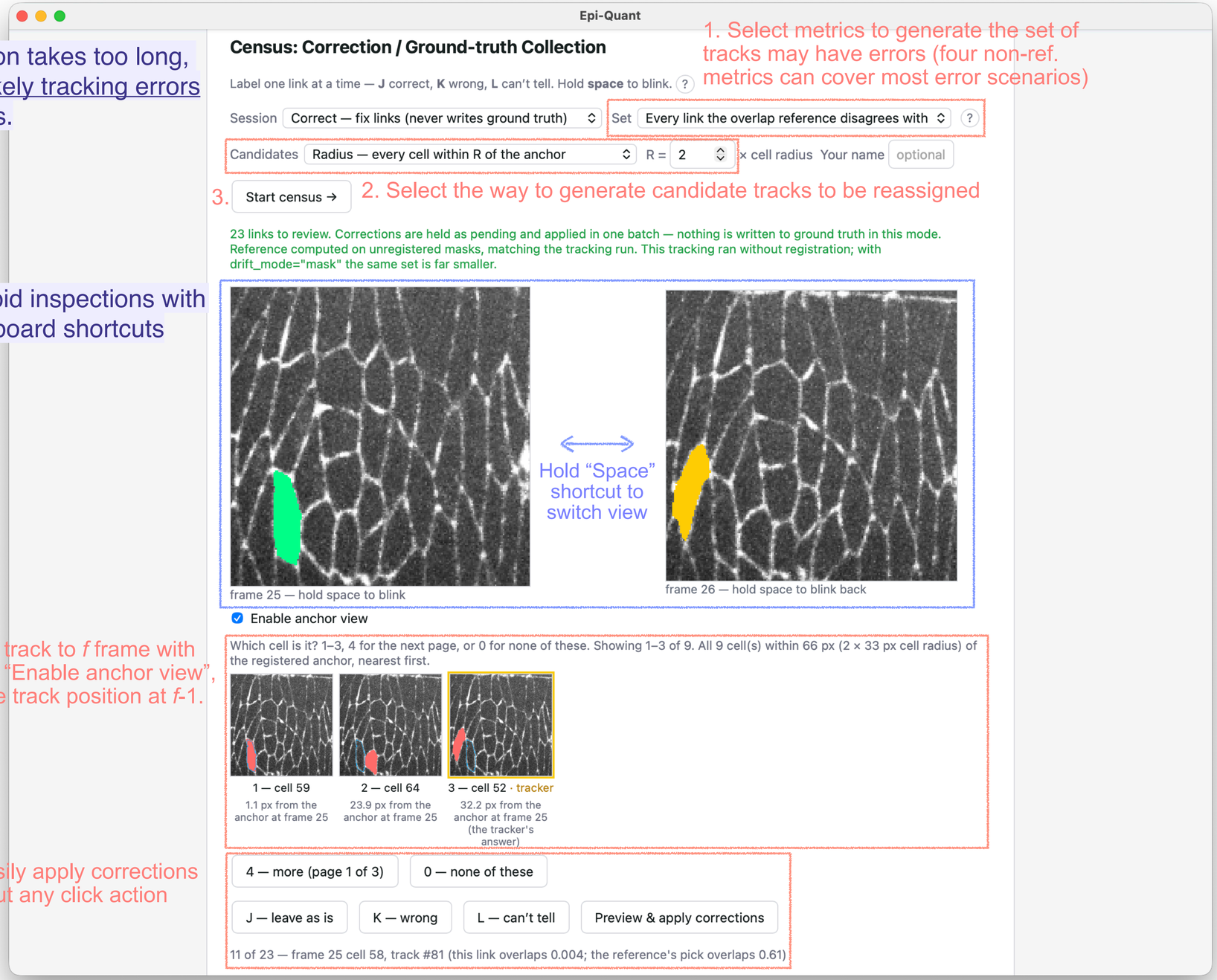
6.6 Tracking Correction with The Aid of Census Module

If the frame-by-frame inspection takes too long, census mode can find most likely tracking errors based on no-reference metrics.

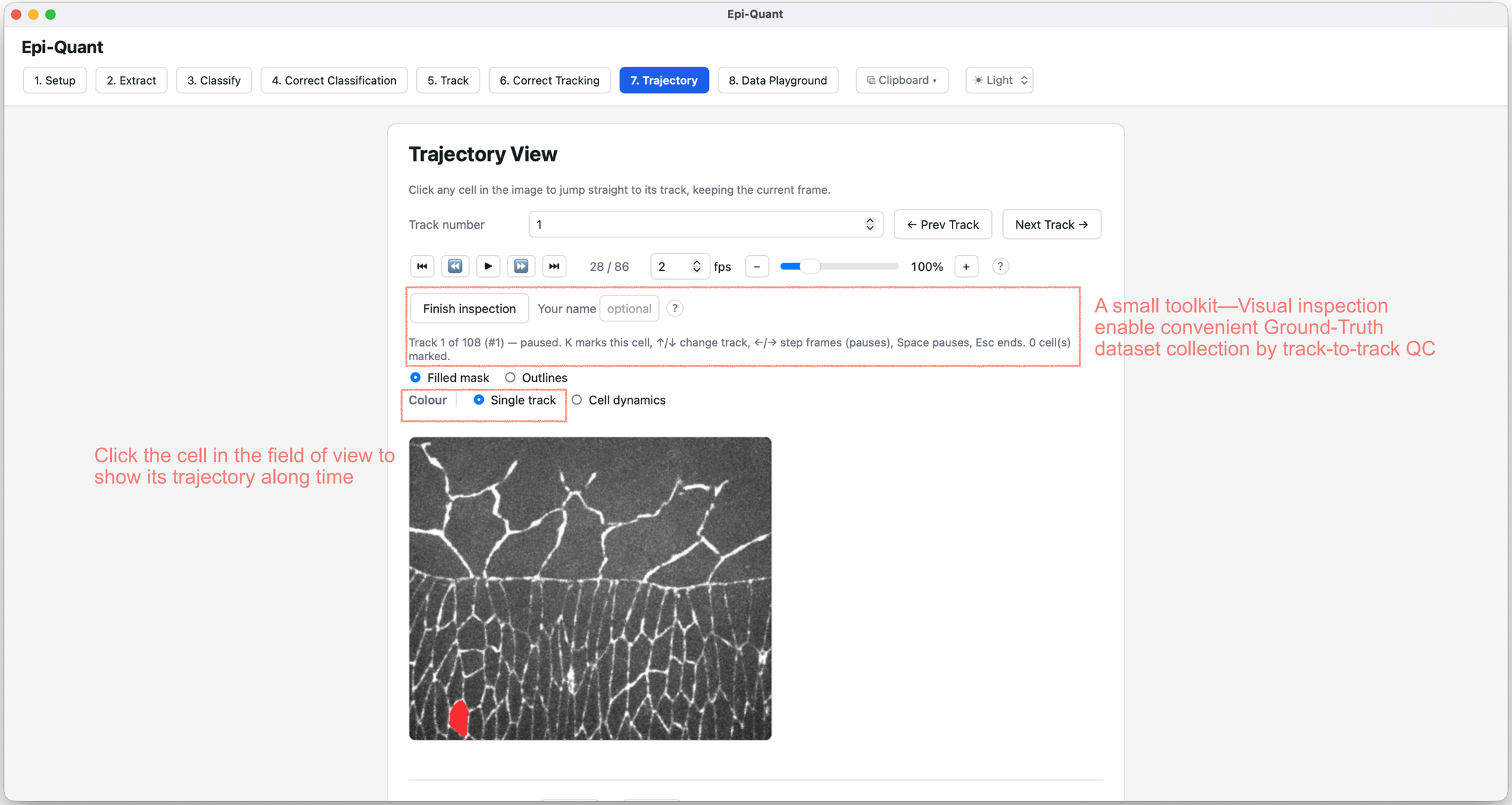
Census mode enable rapid inspections with the aid of easily-use keyboard shortcuts

4. Reassign the track to *f* frame with the visual aid of "Enable anchor view", which shows the track position at *f*-1.

5. Easily apply corrections without any click action



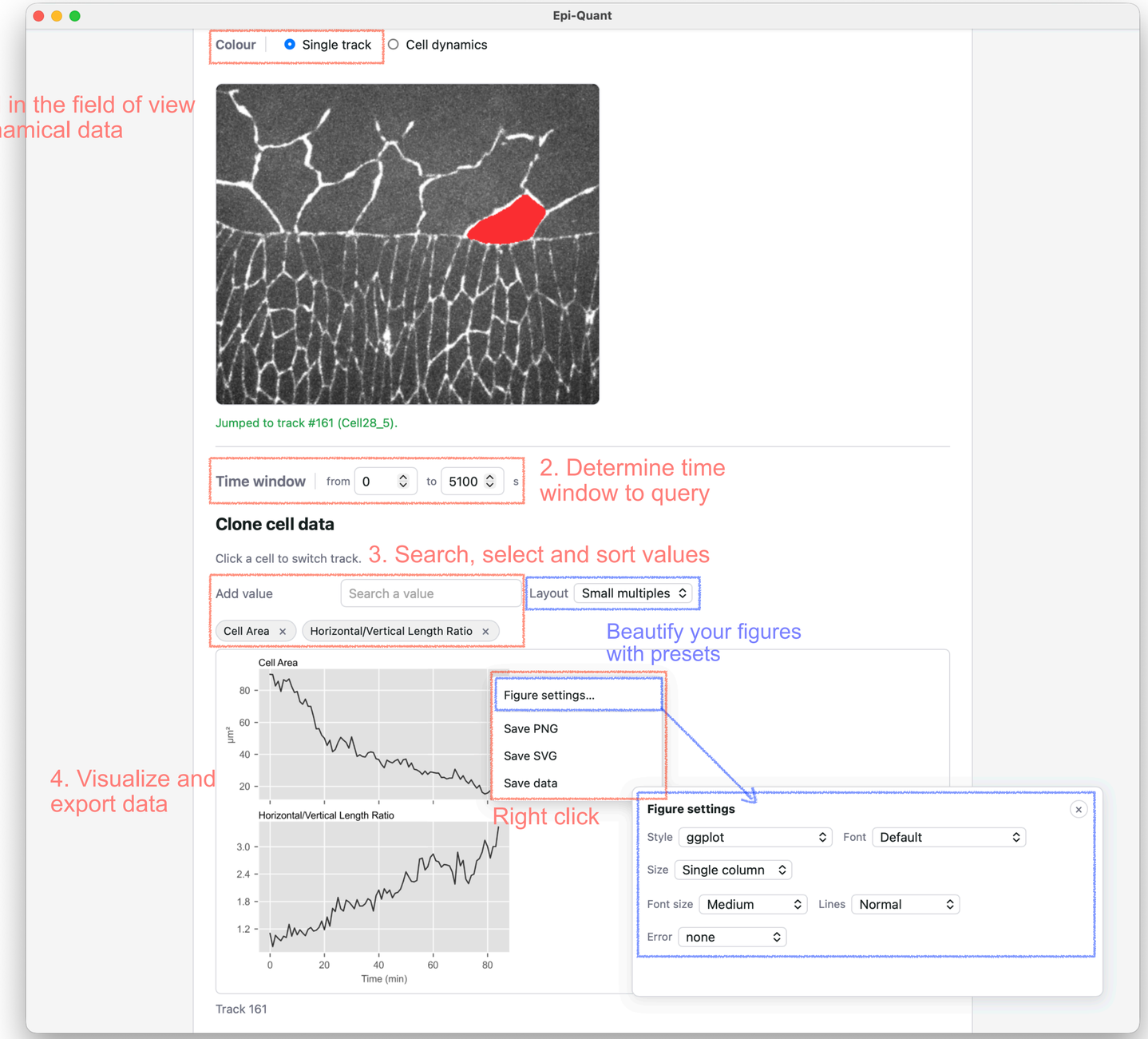
7.1 Overview of Trajectory Module



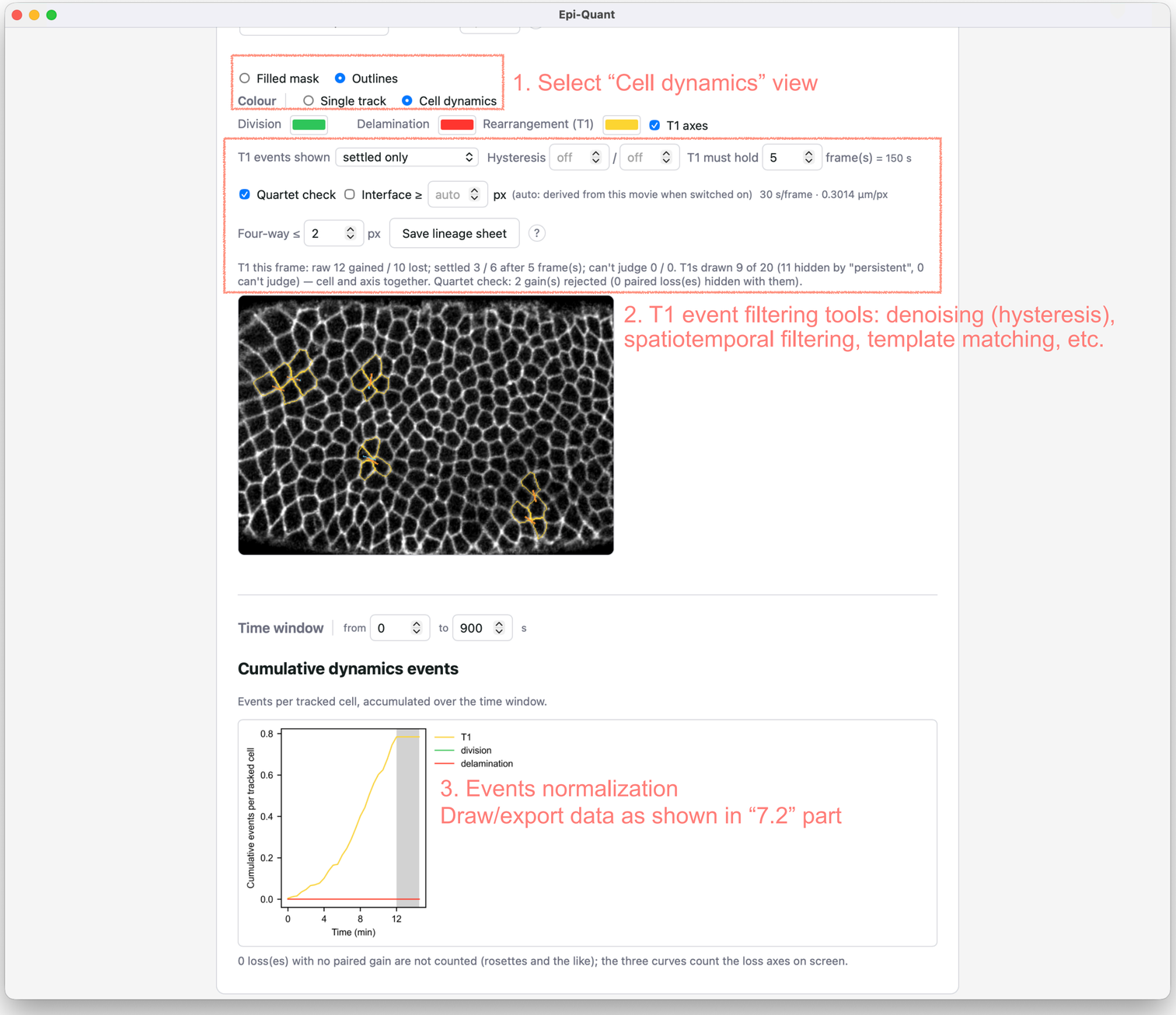
Click the cell in the field of view to show its trajectory along time

A small toolkit—Visual inspection enable convenient Ground-Truth dataset collection by track-to-track QC

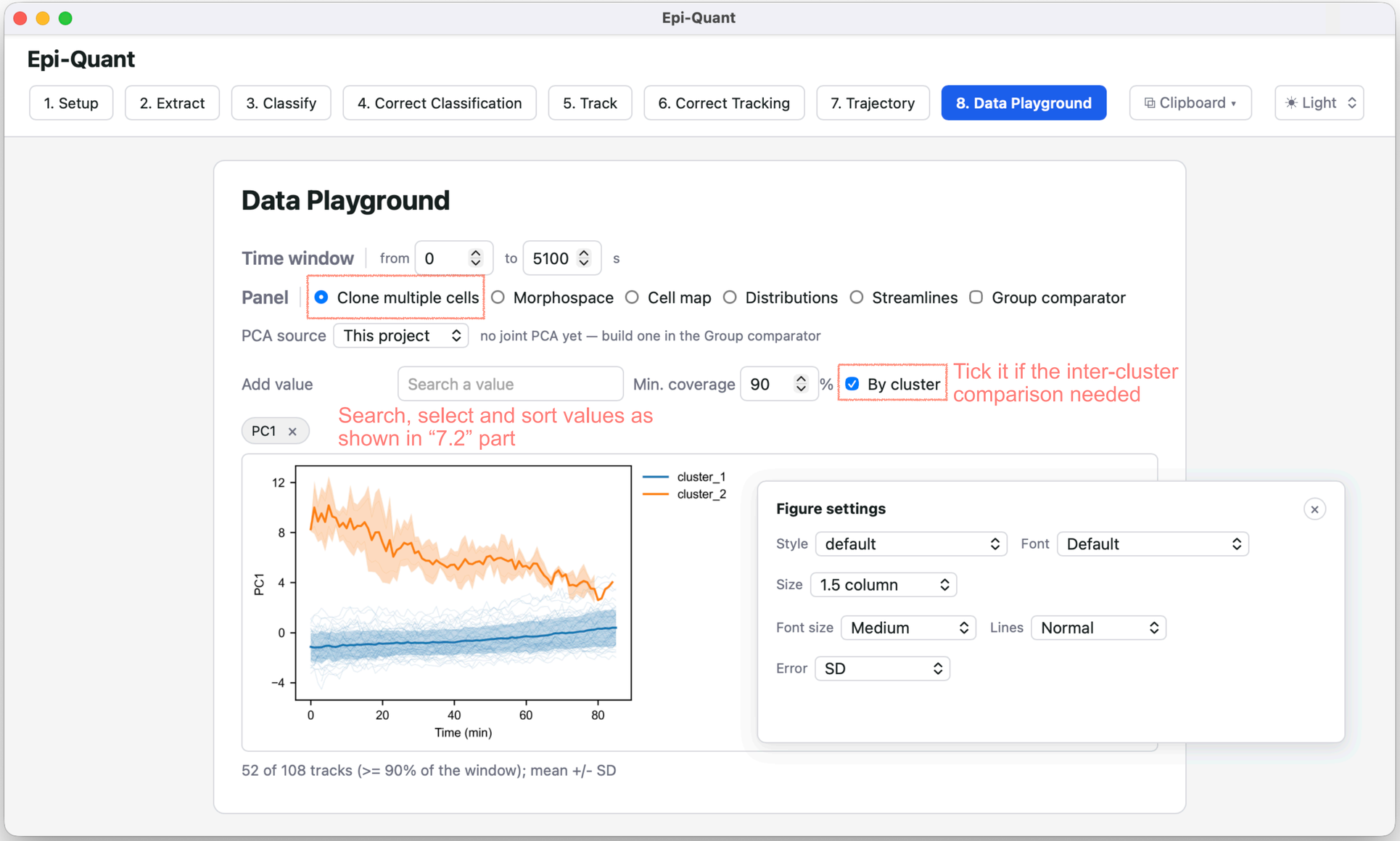
7.2 Single-cell Dynamical Data Visualization And Exportation



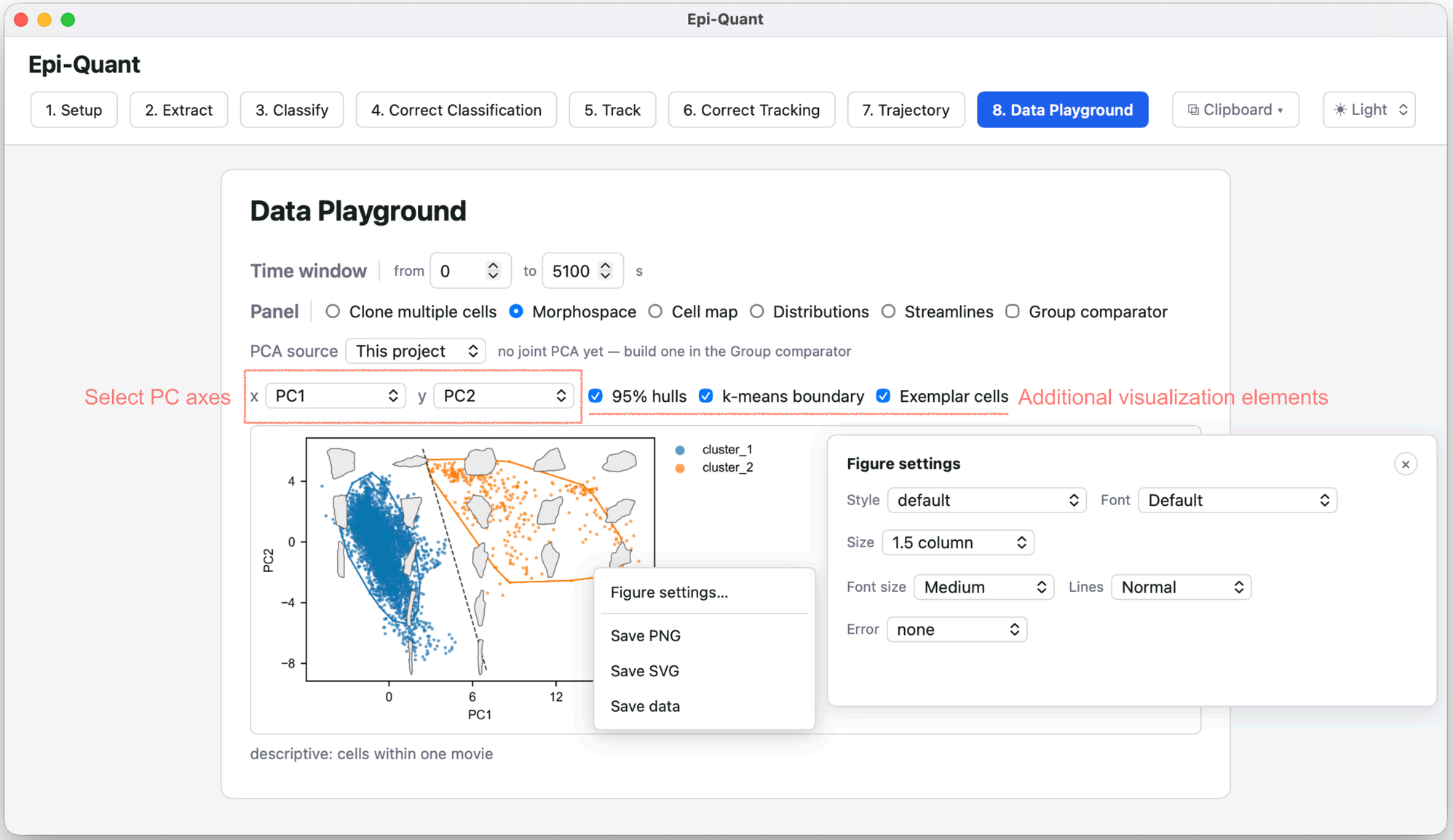
7.3 Dynamical Events Visualization And Exportation



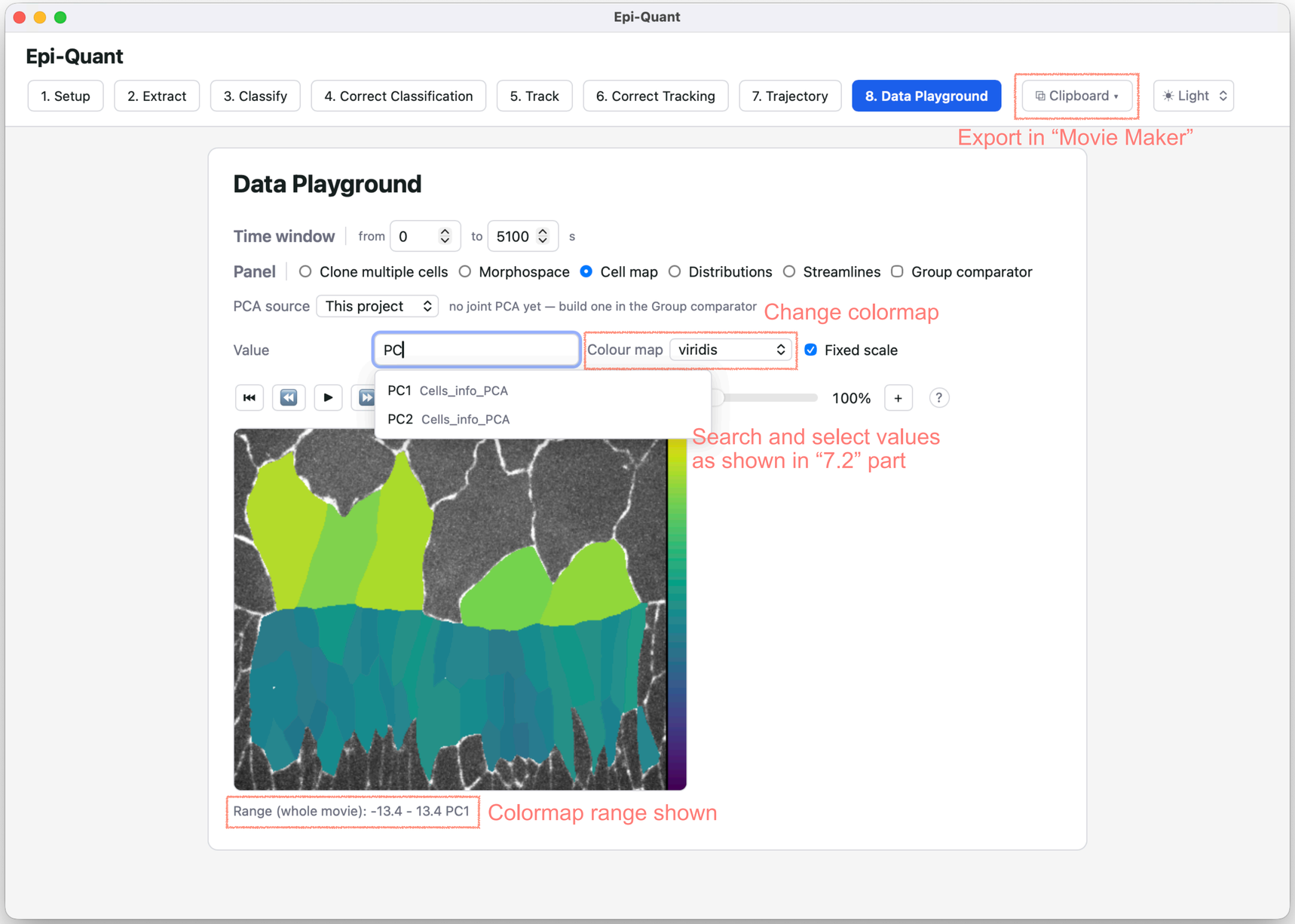
8.1 Data Playground Module — Collective Cell Dynamics Visualization/Exportation



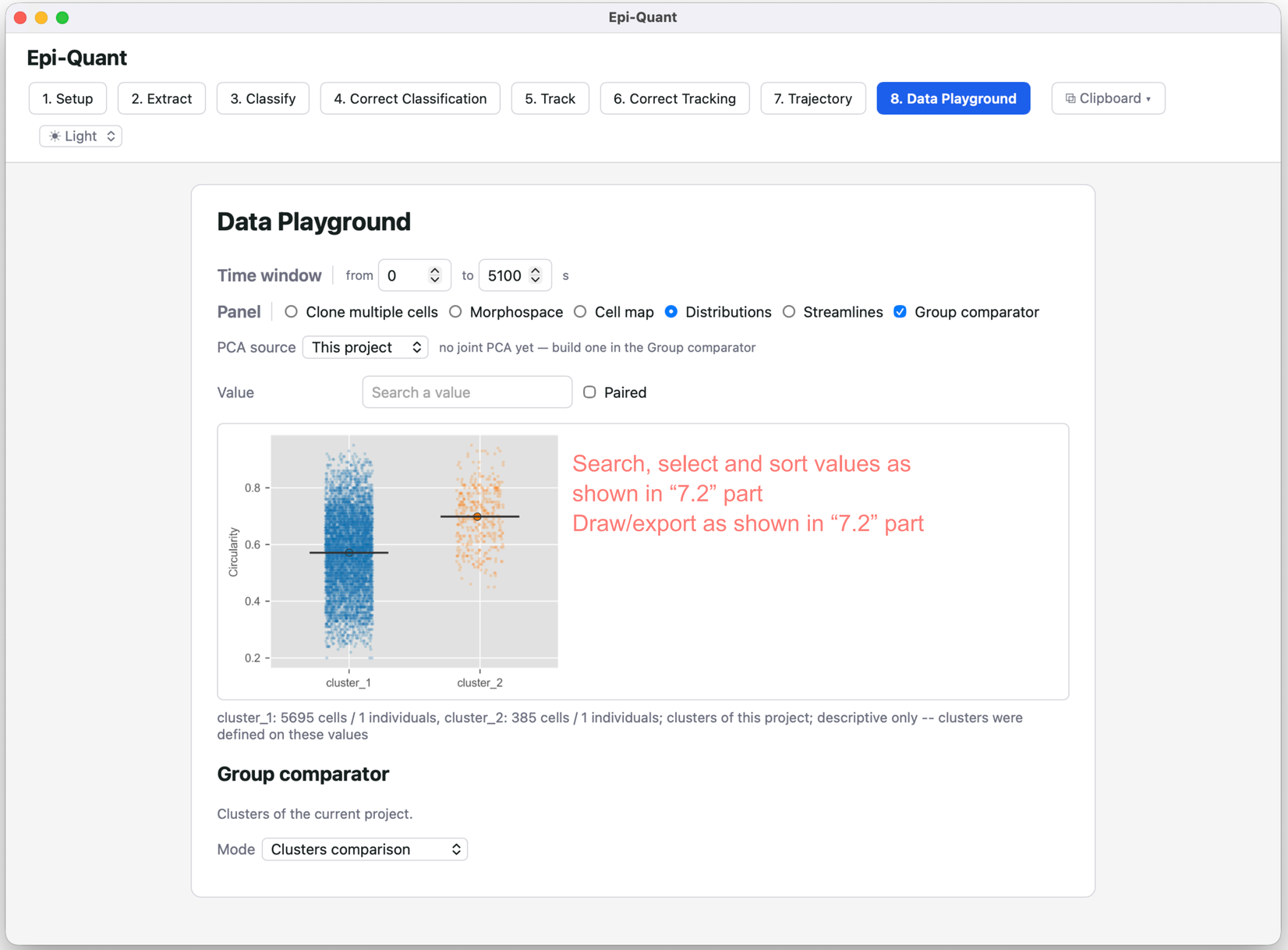
8.2 Data Playground Module — Morphospace



8.3 Data Playground Module — Cell Map Generation



8.4 Data Playground Module — Inter-group Comparison



8.5 Data Playground Module — Cell Streamlines Visualization